- Modularity
- Inheritance
- Polymorphism

13.3.1 Encapsulation

The mechanism by which the data and functions are bound together into a single unit is known as **Encapsulation.** It implements abstraction.

Encapsulation is about binding the data variables and functions together in class. It can also be called **data binding.**

Encapsulation is the most striking feature of a class. The data is not accessible to the outside world, and only those functions which are wrapped in the class can access it. These functions provide the interface between the object's data and the program. **This encapsulation of data from direct access by the program is called data hiding or information hiding.**

13.3.2 Data Abstraction

Abstraction refers to showing only the essential features without revealing background details. Classes use the concept of abstraction to define a list of abstract attributes and function which operate on these attributes. They encapsulate all the essential properties of the object that are to be created. The attributes are called **data members** because they hold information. The functions that operate on these data are called **methods or member function.**

13.3.3 Modularity

Modularity is designing a system that is divided into a set of functional units (named modules) that can be composed into a larger application.

13.3.4 Inheritance

Inheritance is the technique of building new classes **(derived class)** from an existing Class **(base class)**. The most important advantage of inheritance is **code reusability.**

13.3.5 Polymorphism

Polymorphism is the ability of a message or function to be displayed in more than one form.

13.4 Advantages of OOP

Re-usability:

"Write once and use it multiple times" you can achieve this by using class.

Redundancy:

Inheritance is the good feature for data redundancy. If you need a same functionality in multiple class you can write a common class for the same functionality and inherit that class to sub class.

Easy Maintenance:

It is easy to maintain and modify existing code as new objects can be created with small differences to existing ones.

Security:

Using data hiding and abstraction only necessary data will be provided thus maintains the security of data.

13.5 Disadvantages of OOP

Size:

Object Oriented Programs are much larger than other programs.

Effort:

Object Oriented Programs require a lot of work to create.

Speed:

Object Oriented Programs are slower than other programs, because of their size.

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